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Title : Configure, Apply and Verify an Extended Numbered ACL.

Aim : To Configure, Apply and Verify an Extended Numbered ACL.

Configuring the ssh to the routers
1) Set the hostname
2) Set the router enable password
3) Set domain-name
4) Create a local user with privilege level
5) Generate the RSA key
6) Enable the ssh version 2
7) Enable ssh on the VTY lines
8) Test the ssh from the PC

Example ->
```
Router(config)# hostname <your_hostname>
Router(config)# enable secret <enable_key>
R1(config)# ip domain-name <example.com>
R1(config)# username <yourname> privilege 15 secret <passkey>
R1(config)# crypto key generate rsa
How many bits in the modulus [512]: 1024
R1(config)# ip ssh version 2
R1(config)# line vty 0 4
R1(config-line)# login local
R1(config-line)# transport input ssh
R1(config-line)# exit

ssh -l <your_username> <ip_address_of_router_interface>
ssh -l <ip_address_of_router_interface>
```

Access Control List (ACL)
An ACL is a rule-based control mechanism that defines what traffic is allowed or denied to enter or exit a network interface. ACLs check packet headers and then apply actions based on defined rules.

Types of ACL ->
1) Standard ACL
   Filters traffic based only on Source IP address
   Uses ACL numbers 1–99 (IPv4)
   Applied usually closest to the destination
   Example Rule : Allow traffic from 192.168.1.10 only.

2) Extended ACL
   Filters traffic based on:
   Source IP
   Destination IP
   Protocol (TCP, UDP, ICMP, etc.)
   Source/Destination Port
   Uses ACL numbers 100–199
   Example Rule : Allow HTTP traffic from 192.168.1.0/24 to 10.10.10.5.

How ACL Works
When a packet arrives:
Router checks ACL rules top to bottom.
First matched rule is applied.
If no rule matches → Implicit deny (blocked by default).

Benefits of ACL
Increases network security
Limits unnecessary traffic
Controls remote access
Filters routing updates
Provides granular access control

Standard ACL Syntax
access-list <1–99> {permit | deny} <source-ip> <wildcard-mask>

here access-list <number> -> Standard ACL numbers = 1 to 99 Example : access-list 10
permit | deny -> permit = allow, deny = block
<source-ip> -> The address you want to allow/deny Example: 192.168.1.0
<wildcard-mask> -> Opposite of subnet mask, 0 = must match, 255 = can be anything Example wildcard : 0.0.0.255 -> matches a /24 network, 0.0.0.0 → matches a single host

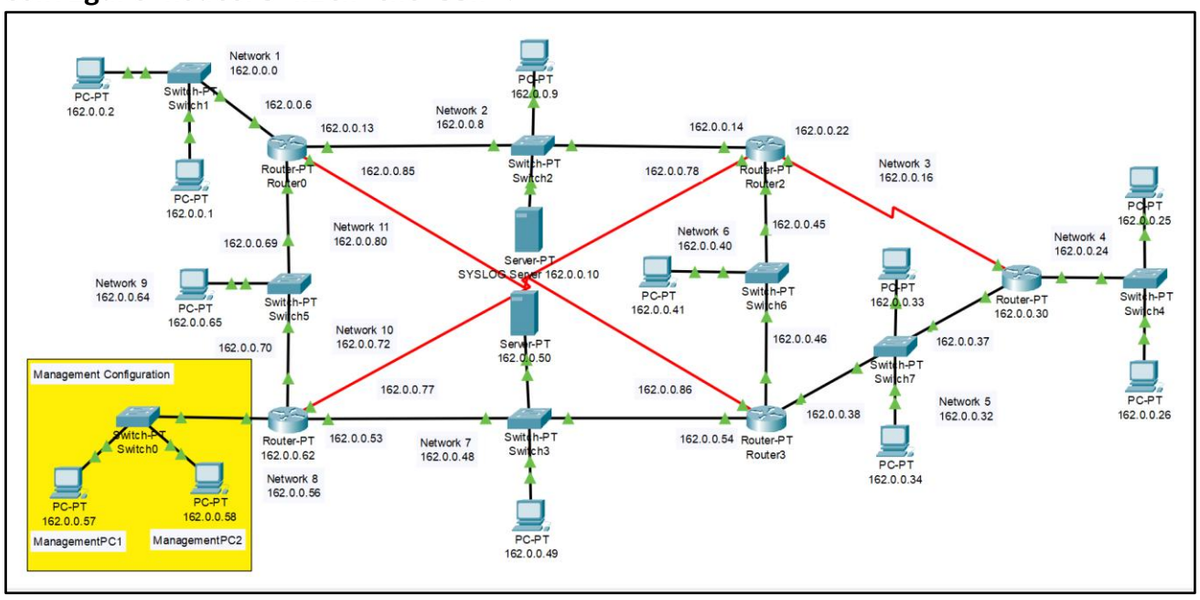
Extended ACL Syntax
access-list <100–199> {permit | deny} <protocol> <source-ip> <source-wildcard> <destination-ip> <destination-wildcard> [operator] [port]

here, access-list <number> -> Extended ACL numbers = 100 to 199
Example : access-list 110
permit | deny -> Allow or block traffic.
<protocol> -> tcp, udp, icmp, ip (all IP protocols)
<source-ip> <source-wildcard> -> Who is sending the packet?
Example : 192.168.1.0 0.0.0.255
destination-ip> <destination-wildcard> -> Who is receiving the packet?
Example : 10.10.10.10 0.0.0.0
Optional: Port number -> Only for TCP/UDP

Operators	Operator Meaning
eq	equal to port
lt	less than port
gt	greater than port range of ports

Example: eq 80 -> HTTP, eq 22 -> SSH, eq 443 -> HTTPS

Configure Routers with the ssh :



Configure ssh Routers -> Router1

```
YashodipRouter1>enable
Password:
YashodipRouter1#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
YashodipRouter1(config)#ip domain-name yashinfotech.com
YashodipRouter1(config)#username admin privilege 15 password admin@123
YashodipRouter1(config)#username yashodip privilege 15 password yashodip@123
YashodipRouter1(config)#crypto key generate rsa
How many bits in the modulus [512]: 2048
YashodipRouter1(config)#ip ssh version 2
YashodipRouter1(config)#line vty 0 4
YashodipRouter1(config-line)#login local
YashodipRouter1(config-line)#transport input ssh
YashodipRouter1(config-line)#exit
```

Router2

```
YashodipRouter2>enable
Password:
YashodipRouter2#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
YashodipRouter2(config)#ip domain-name yashinfotech.com
YashodipRouter2(config)#username admin privilege 15 password admin@123
YashodipRouter2(config)#username vaibhavi privilege 15 password vaibhavi@123
YashodipRouter2(config)#crypto key generate rsa
The name for the keys will be: YashodipRouter2.yashinfotech.com
How many bits in the modulus [512]: 2048
YashodipRouter2(config)#ip ssh version 2
YashodipRouter2(config)#line vty 0 4
YashodipRouter2(config-line)#login local
YashodipRouter2(config-line)#transport input ssh
YashodipRouter2(config-line)#exit
```

Router3

```
YashodipRouter3>enable
Password:
YashodipRouter3#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
YashodipRouter3(config)#ip domain-name yashinfotech.com
YashodipRouter3(config)#username admin privilege 15 password admin@123
YashodipRouter3(config)#username lajari privilege 15 password lajari@123
YashodipRouter3(config)#crypto key generate rsa
How many bits in the modulus [512]: 2048
YashodipRouter3(config)#ip ssh version 2
YashodipRouter3(config)#line vty 0 4
YashodipRouter3(config-line)#login local
YashodipRouter3(config-line)#transport input ssh
YashodipRouter3(config-line)#exit
```

Router4

```
YashodipRouter4>enable
Password:
YashodipRouter4#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
YashodipRouter4(config)#ip domain-name yashinfotech.com
YashodipRouter4(config)#username admin privilege 15 password admin@123
YashodipRouter4(config)#username shruti privilege 15 password shruti@123
YashodipRouter4(config)#crypto key generate rsa
How many bits in the modulus [512]: 2048
YashodipRouter4(config)#ip ssh version 2
YashodipRouter4(config)#line vty 0 4
YashodipRouter4(config-line)#login local
YashodipRouter4(config-line)#transport input ssh
YashodipRouter4(config-line)#exit
```

Router5

```
YashodipRouter5>enable
Password:
YashodipRouter5#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
YashodipRouter5(config)#ip domain-name yashinfotech.com
YashodipRouter5(config)#username admin privilege 15 password admin@123
YashodipRouter5(config)#username sonia privilege 15 password sonia@123
YashodipRouter5(config)#crypto key generate rsa
How many bits in the modulus [512]: 2048
YashodipRouter5(config)#ip ssh version 2
YashodipRouter5(config)#line vty 0 4
YashodipRouter5(config-line)#login local
YashodipRouter5(config-line)#transport input ssh
YashodipRouter5(config-line)#exit
```

Router	User1	Password1	User2	Password2
R1	admin	admin@123	yashodip	yashodip@123
R2	admin	admin@123	vaibhavi	vaibhavi@123
R3	admin	admin@123	lajari	lajari@123
R4	admin	admin@123	shruti	shruti@123
R5	admin	admin@123	sonia	sonia@123

Accessing the ssh using PC ->
Packet Tracer PC Command Line 1.0
C:\>ssh -l admin 162.0.0.30
Password:
YashodipRouter3#show clock
*1:12:26.673 UTC Mon Mar 1 1993

Standard Access Control List
Case 1 : allow only one specific PC
Allow 162.0.0.33 into the network 4 162.0.0.24 and denied for the others

```
YashodipRouter3>enable
Password:
YashodipRouter3#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
YashodipRouter3(config)#access-list 1 permit 162.0.0.33 0.0.0.0
YashodipRouter3(config)#interface fa0/0
YashodipRouter3(config-if)#ip access-group 1 out
YashodipRouter3(config-if)#exit

C:\>ping 162.0.0.25    //ping command from the pc 162.0.0.33
Pinging 162.0.0.25 with 32 bytes of data:
Reply from 162.0.0.25: bytes=32 time=11ms TTL=127
Reply from 162.0.0.25: bytes=32 time=2ms TTL=127
Reply from 162.0.0.25: bytes=32 time<1ms TTL=127
Reply from 162.0.0.25: bytes=32 time=1ms TTL=127
Ping statistics for 162.0.0.25:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 11ms, Average = 3ms

C:\>ping 162.0.0.25    //ping command from the pc 162.0.0.41
Pinging 162.0.0.25 with 32 bytes of data:
Request timed out.
Reply from 162.0.0.37: Destination host unreachable.
Reply from 162.0.0.37: Destination host unreachable.
Reply from 162.0.0.37: Destination host unreachable.
Ping statistics for 162.0.0.25:
Packets: Sent = 4, Received = 0, Lost = 4 100% loss
```

Case 2 : allow entire network
Allow the access for the networks 162.0.0.0 and 162.0.0.56 in the network 162.0.0.24

```
YashodipRouter3>enable
Password:
YashodipRouter3#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
YashodipRouter3(config)#access-list 2 permit 162.0.0.0 0.0.0.7
YashodipRouter3(config)#access-list 2 permit 162.0.0.56 0.0.0.7
YashodipRouter3(config)#interface fa0/0
YashodipRouter3(config-if)#ip access-group 2 out
YashodipRouter3(config-if)#exit
```

Case 3 : deny one host and allow everyone else
Deny the 162.0.0.41 and allow for all host to the router1 to access the network 162.0.0.0

```
YashodipRouter1>enable
Password:
YashodipRouter1#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
YashodipRouter1(config)#access-list 3 deny 162.0.0.41 0.0.0.0
YashodipRouter1(config)#access-list 3 permit any
YashodipRouter1(config)#interface fa0/0
YashodipRouter1(config-if)#ip access-group 3 out
YashodipRouter1(config-if)#exit

C:\>ping 162.0.0.1     //ping from 162.0.0.41
Pinging 162.0.0.1 with 32 bytes of data:
Reply from 162.0.0.13: Destination host unreachable.
Reply from 162.0.0.13: Destination host unreachable.
Reply from 162.0.0.13: Destination host unreachable.
Reply from 162.0.0.13: Destination host unreachable.
Ping statistics for 162.0.0.1:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)
```

Case 4 : allow only the even ip from the net 162.0.0.0 at router 5 in the network of the 162.0.0.56

```
YashodipRouter5>enable
Password:
YashodipRouter5#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
YashodipRouter5(config)#access-list 4 permit 162.0.0.0 0.0.0.254
YashodipRouter5(config)#interface fa1/0
YashodipRouter5(config-if)#ip access-group 4 out
YashodipRouter5(config-if)#exit

C:\>ping 162.0.0.57 //from pc 162.0.0.2 (even)
Ping statistics for 162.0.0.57:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 11ms, Average = 3ms

C:\>ping 162.0.0.57 //from pc 162.0.0.1 (odd)
Ping statistics for 162.0.0.57:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Case 5 : block whole subnet network 162.0.0.16

```
Example router(config)#access-list 5 deny 162.0.0.16 0.0.0.7
        router(config)#access-list 5 permit any

YashodipRouter1#show access-lists  // for showing the access list
Standard IP access list 3
    10 deny host 162.0.0.41 (26 match(es))
    20 permit any (15 match(es))
Extended IP access list 101
    10 permit tcp 162.0.0.0 0.0.0.7 any eq www (24 match(es))
Standard IP access list 9
    10 permit 162.0.0.0 0.0.0.7 (4 match(es))
```

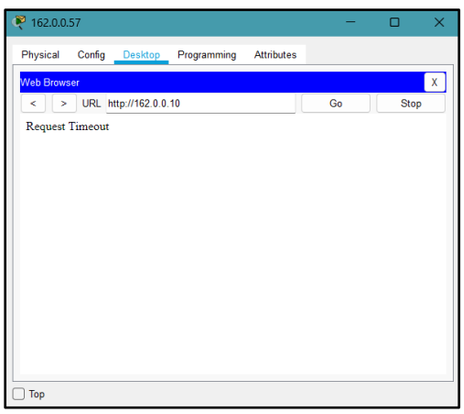
Remove the access-list

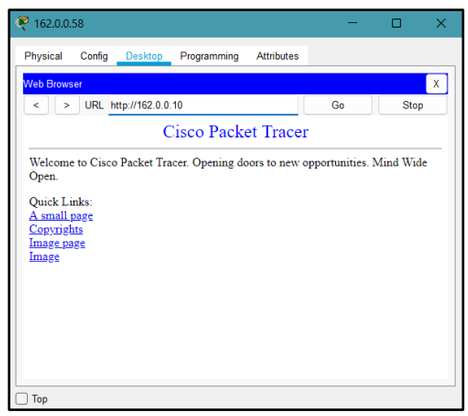
```
YashodipRouter1(config)#no access-list 3
YashodipRouter1(config)#no access-list 9
```

Extended Access Control List

Case 1 : deny only http (port 80) from host 162.0.0.57 and allow for all other for the host server 162.0.0.10

```
YashodipRouter5>en
Password:
YashodipRouter5#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
R5(config)#access-list 121 deny tcp host 162.0.0.57 host 162.0.0.10 eq 80
R5(config)#access-list 121 deny tcp host 162.0.0.57 host 162.0.0.10 eq 443
YashodipRouter5(config)#access-list 121 permit tcp any any eq 80
YashodipRouter5(config)#access-list 121 permit tcp any any eq 443
YashodipRouter5(config)#int fa1/0
YashodipRouter5(config-if)#ip access-group 121 in
YashodipRouter5(config-if)#exit
```





Case 2 : allow the ssh for only 162.0.0.57 and 162.0.0.58 for the router 162.0.0.62

```
YashodipRouter5>en
Password:
YashodipRouter5#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
R5(config)#access-list 122 permit tcp host 162.0.0.57 host 162.0.0.62 eq 22
R5(config)#access-list 122 permit tcp host 162.0.0.58 host 162.0.0.62 eq 22
YashodipRouter5(config)#int fa1/0
YashodipRouter5(config-if)#ip access-group 122 in
YashodipRouter5(config-if)#exit
YashodipRouter5(config)#int fa0/0
YashodipRouter5(config-if)#ip access-group 122 in
YashodipRouter5(config-if)#exit
YashodipRouter5(config)#int fa6/0
YashodipRouter5(config-if)#ip access-group 122 in
YashodipRouter5(config-if)#exit
YashodipRouter5(config)#int se3/0
YashodipRouter5(config-if)#ip access-group 122 in
YashodipRouter5(config-if)#exit
```

```
Access the ssh service from the managementPc1  //162.0.0.57
C:\>ssh -l sonia 162.0.0.62
Password:
YashodipRouter5#show clock
*0:19:44.336 UTC Mon Mar 1 1993

Access the ssh service from the managementPc2  //162.0.0.58
C:\>ssh -l sonia 162.0.0.62
Password:
YashodipRouter5#show clock
*0:19:48.336 UTC Mon Mar 1 1993

Access the ssh service from the 162.0.0.33
Packet Tracer PC Command Line 1.0
C:\>ssh -l sonia 162.0.0.62
% Connection timed out; remote host not responding
```